FACE RECOGNITION- BASED ATTENDANCE SYSTEM

#Project Report#

Submitted by

Name: Khushi Singh

Registration Number: 25BCE11223

Course: CSA2001

(Fundamentals of AI & ML)

Introduction

Attendance management is a fundamental and critical administrative task in educational institutions and corporate environments. This process depends on manual methods like calling out names or circulating paper sign-in sheets. These systems are generally time consuming, disrupt the flow of lectures or work, and are highly inaccurate and unethical practices, most notably proxy attendance (where one person marks another absent individual as present). So, to address these inefficiencies, this project proposes an advanced solution to the same: “the Face Recognition-Based Attendance System.”

This system uses computer vision and machine learning technology to recognise people in a live video stream and automatically mark their presence. Video input is captured, human faces are detected, the identity of the detected face is determined by comparing it with a database of pre-registered facial encodings, and then attendance is then recorded with a timestamp.

This system's successful implementation shows the practical application of algorithms like Haar Cascades or Histogram of Orientated Gradients for detection and FaceNet for extremely accurate face recognition. The primary tools used in the project's Python construction are the OpenCV and face recognition libraries, which ensures a dependable and efficient solution.

The overall goal is to provide a reliable, efficient, and non-intrusive mechanism for attendance record-keeping, saving valuable institutional time and improving the attendance data.

PROJECT OBJECTIVE

The primary objectives of this project are:

1. Academic Application: To apply core concepts and techniques of Computer Vision and Machine Learning to a meaningful, real-world application.

2. Technical Solution: To design and implement a robust, automated, and non-intrusive system which is capable of recording attendance in real-time accurately.

By achieving this, the project aims to revolutionize the method of attendance taking by:

- Eliminating Time Wastage: Automating the process to reduce the time consumed by manual roll calls.

- Enhancing Accuracy and Reliability: Using biometric verification to minimize human error and completely remove the possibility of proxy attendance which is a major issue.

- Digitizing Records: Providing a secure, centralized, and easily searchable digital database for attendance records and report generation.

Scope and Requirements

The scope defines the boundaries, features, and performance constraints of the "Face Recognition-Based Attendance System." This section is about the specific functions to be implemented and the quality attributes the system is designed to meet.

In-Scope:

• Real-Time Attendance Marking: The system will capture live video from a standard webcam, detect faces, and automatically mark the attendance of recognized, pre registered users with a date and time stamp.

• User Enrolment and Training: An administrator interface will allow for capturing multiple facial images per user and generating 128-dimensional face encodings for the recognition model's database.

• Database Management: The system will store user profile data (Name, ID, Encodings) and manage the daily attendance log records.

• Report Generation: The system will allow to view, filter, and export attendance reports for administrative use.

Out-of-Scope :

• Advanced Anti-Spoofing: This project scope does not include complex anti-spoofing mechanisms (e.g., blink detection or 3D mapping) to verify a live person v/s a photograph.

• Cloud Deployment: The system will be implemented as a local, standalone applications and will not be deployed to a cloud platform for web-based access.

• Multi-Camera Support: The implementation is limited to processing video input from a single, given camera source at any given time.

Functional Requirements

Functional Requirements define the capabilities the Face Recognition-Based Attendance System must perform. The system is designed around the core operational workflow of registration, recognition, and reporting.

1. User Enrolment and Training Module (Registration): This module manages the process of bringing new users into the system and preparing the face recognition model.

Feature	Description	Input	Output
Registration of User	Allows administrator to enter new user's unique details (e.g., student ID, employee number) and name		
Image Capture	Captures multiple live facial images (e.g., 20 to 50 samples) of the user under varied situations (slight pose changes, expressions) to ensure strong training data		
Encoding Generation	Processes the captured images to generate a single, unique, 128 dimensional face encoding for the user		

2. Real-Time Attendance Marking Module (Recognition): It is the primary operational module responsible for automating the attendance process.

Feature	Description	Input	Output
Video Stream capture	Starts and maintains a connection to the system's webcam, simultaneously capturing video frames in real time	Camera Hardware Input	Continuous stream of video frames
Face detection	For each frame, the system must identify the location and size of all human faces present using an algorithm like HOG or Haar Cascades	Video Frame	Bounding box coordinates for each and every detected face
Face recognition	Compares the live face encoding against all others whose data is available, stored encodings in the database to identify the closest match	Live Face Encoding, Database of Known Encodings	The ID and Name of the recognized user (if it's a match)
Attendance logging	If the match is successful, the user is marked as 'Present' in the attendance database. Mechanism must prevent marking the same user multiple times present within a short interval (e.g., 10 minutes)	Recognized User ID, System Time	New record added to the Attendance Log (User ID, Date, Time, Status: Present)

User feedback	Displays the recognized user's name on the screen	Recognized User Name	Visual display of the recognized name on the live video stream

3. Data Reporting and Administration Module (Reporting): This module handles the retrieval, presentation, and management of the attendance data.

Features	Description	Input	Output
Secured Login	It requires valid administrator credentials to access sensitive features such as user enrolment and report generation	Username and Password	Access granted or denied to the admin interface
Attendance Query	Allows administrators to filter and view attendance records based on specific criteria	Query Parameters (Date, ID)	Tabular display of filtered attendance records on the GUI
Data Export	Generates and saves the queried attendance records into a standard, machine readable file format for external use and archival	Export Command, File Path	Attendance report file created in CSV or Excel format

Non Functional Requirements

Non-functional requirements (NFRs) mean the criteria for judging the operation of the system, rather than just its specific behaviours. They impose constraints on the design and implementation to make sure that the system meets performance, reliability, security, and usability standards and criteria.

ID	Category	Description	Required Criterion
NRF 1	Performance	System must process video frames, perform face recognition, and log attendance quickly to ensure a better experience and prevent any hinder at the entry point	The system must mark attendance with a maximum latency of 3 seconds from the moment the user's face is clearly visible to the camera.
NRF 2	Accuracy & Reliability	The core face recognition algorithm must be highly accurate to minimize misidentifications or failure to identify registered users	The face recognition module must maintain an identification accuracy of at least 90%. The system must run continuously for four hours without crashing
NRF 3	Security	Sensitive enrolment and reporting features must be protected, and the integrity of the attendance data must be maintained	Administrator login and password protection are required for access to the administrative dashboard (enrolment, reporting). Face

			encodings will be kept as non reversible, secure vector data
NRF 4	Usability	The system must be strong enough to function dependably despite minor, real-world variations in the operating system	The recognition system must take variations in ambient lighting and slight head movements / pose changes without failure to recognize a registered user
NRF 5	Maintainability	The software should be easily adaptable to updates, bug f ixes, and future feature enhancements	All significant functions and logic blocks must have thorough inline code comments, and the project must employ a modular design with distinct classes for the camera, database, and recognition

TECHNICAL REQUIREMENTS

Describes the particular hardware and software elements that are necessary to set up, run, and maintain the Face Recognition-Based Attendance System.

1. **Hardware Requirements:** These are the physical components necessary to operate the attendance system.

• Processor (CPU): A multi-core processor (Intel Core i3/AMD Ryzen 3) is generally preferred.

• Memory (RAM): A minimum of 4 GB RAM is required, but 8 GB or higher are highly advised for smooth implementation of computer vision libraries and Python environments.

• Camera: A Webcam or IP Camera with a resolution of at least 720p (HD) is important to capture clear facial images for both enrolment and recognition purpose.

• Storage: Enough hard drive space (such as a 256 GB SSD) is needed for the operating system, software, training photos, and attendance records.

2. **Software Requirements:** Following are the software and tools required:

   • Operating System: Windows, macOS, or Linux (Ubuntu is highly recommended for development with Python).
   • Programming Language: Python 3.x (latest stable version recommended) is the core language for implementation.
   • Development Environment: A Python Integrated Development Environment like PyCharm, VS Code, or Spyder.
   • Database: A simple database system like SQLite or MySQL is used to store user data, face encodings, and the final attendance logs.

3. **Implementation Algorithms:** Following are the key algorithms :

   • Face Detection: Haar Cascades or Histogram of Oriented Gradients.
   • Face Recognition/Feature Extraction: FaceNet (implemented via the face recognition library) to convert faces into embeddings.
   • Matching/Verification: Euclidean Distance or Cosine Similarity is used to compare the distance between the live face embedding and the stored embeddings to get a match.

System Architecture

This project is structured into five different, interconnected layers: Input, Processing, Database, Recognition, and Output. This modular design ensures to remove all the concerns and fulfilling the aim of the project.

1. **Input Layer (Data Acquisition)**

   Webcam/Camera: It is the primary hardware component that constantly captures the real-time video stream of the room or the area where attendance is being taken.

   Administrator Interface: Provides the input point for registering new users and for getting report commands.
2. **Processing Layer (Computer Vision)** This layer handles the core image manipulation and feature extraction.

   Frame Pre-processing: Captured frames are resized to optimize speed and then converted to a suitable colour space (like, RGB) for the recognition algorithm.

   Face Detection: The algorithm scans the frame to locate and draw bounding boxes around each and every face present in the current view.

   Feature Extraction (Encoding): The detected face area is fed to the FaceNet model, which generates a unique 128-dimensional vector.
3. **Database Layer:** This layer is the data storage for the system.

   Training Database: Stores the pre-computed face embeddings of all the registered users, linked to their name and ID. It's the reference of identification of the person or the user.

   User Profiles: Stores metadata associated with each user (e.g., Name, ID, Course/Department).

   Attendance Log: A table (such as, MySQL) that records the date, time, and status of every recognized person.
4. **Recognition & Decision Layer:** Identity and attendance are determined by the fundamental logic.

   Comparison Algorithm: A distance metric is used to compare the live face embedding to all stored known encodings.

   Verification: If the distance to the closest known face is above the required threshold, the identity is verified, and the user is marked as Present. If no close match is found, the face is labelled as Unknown or Absent.

   Attendance Update Logic: Manages the logging process to make sure that a user is only marked present once per session or within a specified time period.

5. **Output Layer (User Interface):** This layer provides feedback to the users and the administrator.
   Real-time Visual Feedback: It displays the live video stream with bounded boxes and the recognized user's Name/ID shown below on the screen.
   Admin/Reporting Interface (GUI): It is a user-friendly graphical interface for accessing the enrolment module, triggering the Attendance Module, and generating exportable data reports.

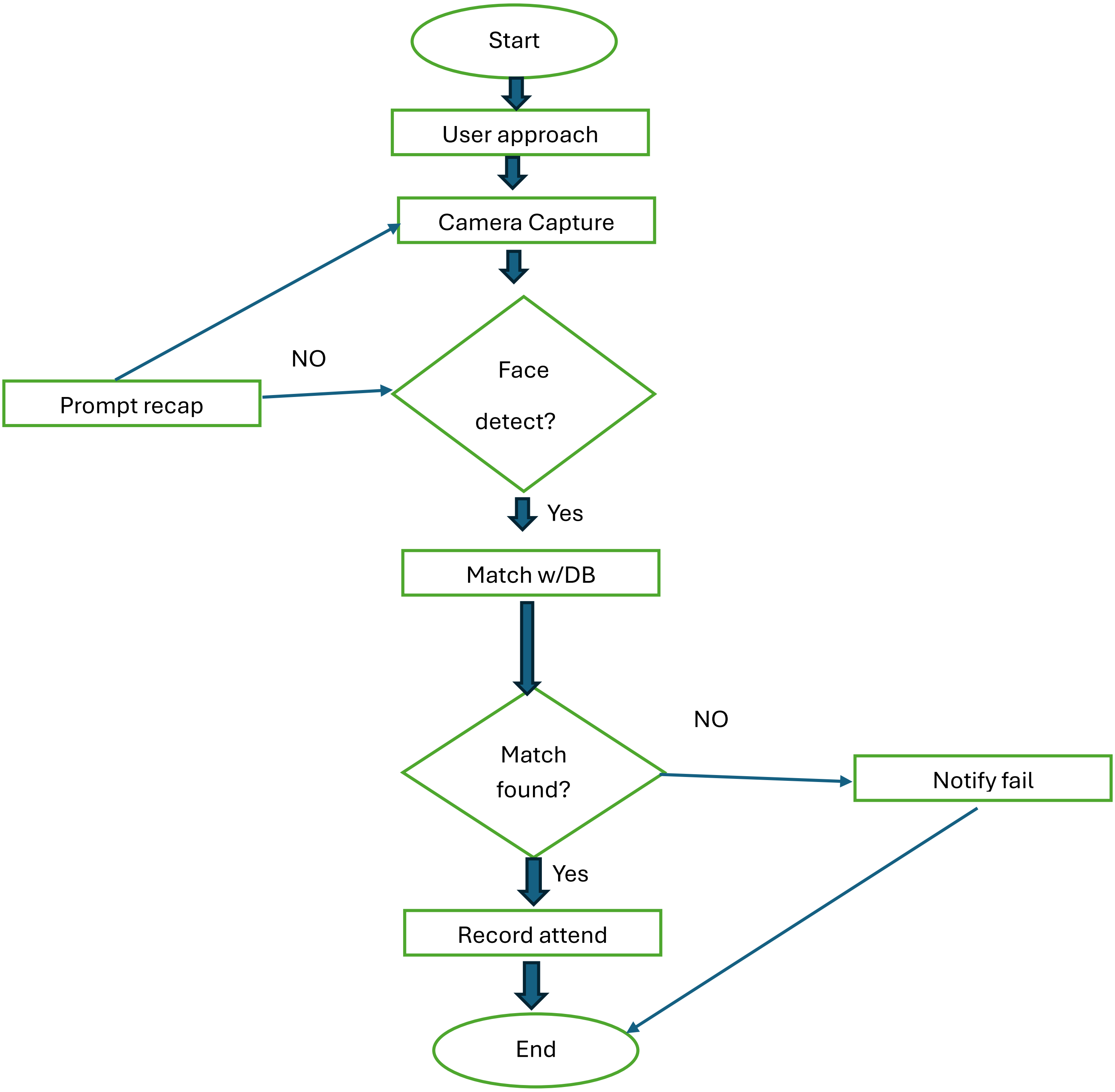
DESIGN DIAGRAMS

Workflow diagram:

Steps:

1. Start System: initiates the attendance process.
2. Initialize Camera: system activates the webcam and begins capturing video frames.
3. Capture Frame: A single image frame is taken from the video stream.
4. Detect Faces: The frame is processed to locate and recognize all faces present.
5. Loop (Face Check): The system repeats through each detected face.
6. Extract Encoding: unique 128-dimensional face encoding is generated for the chosen or given face.
7. Compare Encodings: The live encoding is compared against the Training Database (Known Encodings).
8. Match Found?
   - (Yes): The user's ID is valid hence marked “Present”.
   - (No): The face is temporarily labelled "Unknown."
9. Attendance Marked? Checks if the identified user has already been marked present within the current time period.
   - (No): The attendance log is updated, marking the user as Present with a timestamp.
10. Display Feedback: The system displays the user's name on the screen (or "Unknown").
11. Continue? The loop continues to the next face or returns to Capture Frame until the Administrator issues the Stop System command.

Face Recognition Attendance System



Use Case Diagram Components

- **Actors**
  - User (Student/Employee)
  - Admin
- **User Use Cases**
  - Approach camera for attendance
  - Get attendance recorded via face recognition
  - Receive attendance confirmation or notification
- **Admin Use Cases**
  - Manage user database (add/update/delete users)
  - View and generate attendance records
  - Configure system settings (camera, alerts, database)
- **Relationships**
  - The User interacts with the attendance recording and notification use cases.
  - The admin maintains the user database and monitors attendance logs.

Sequence Diagram

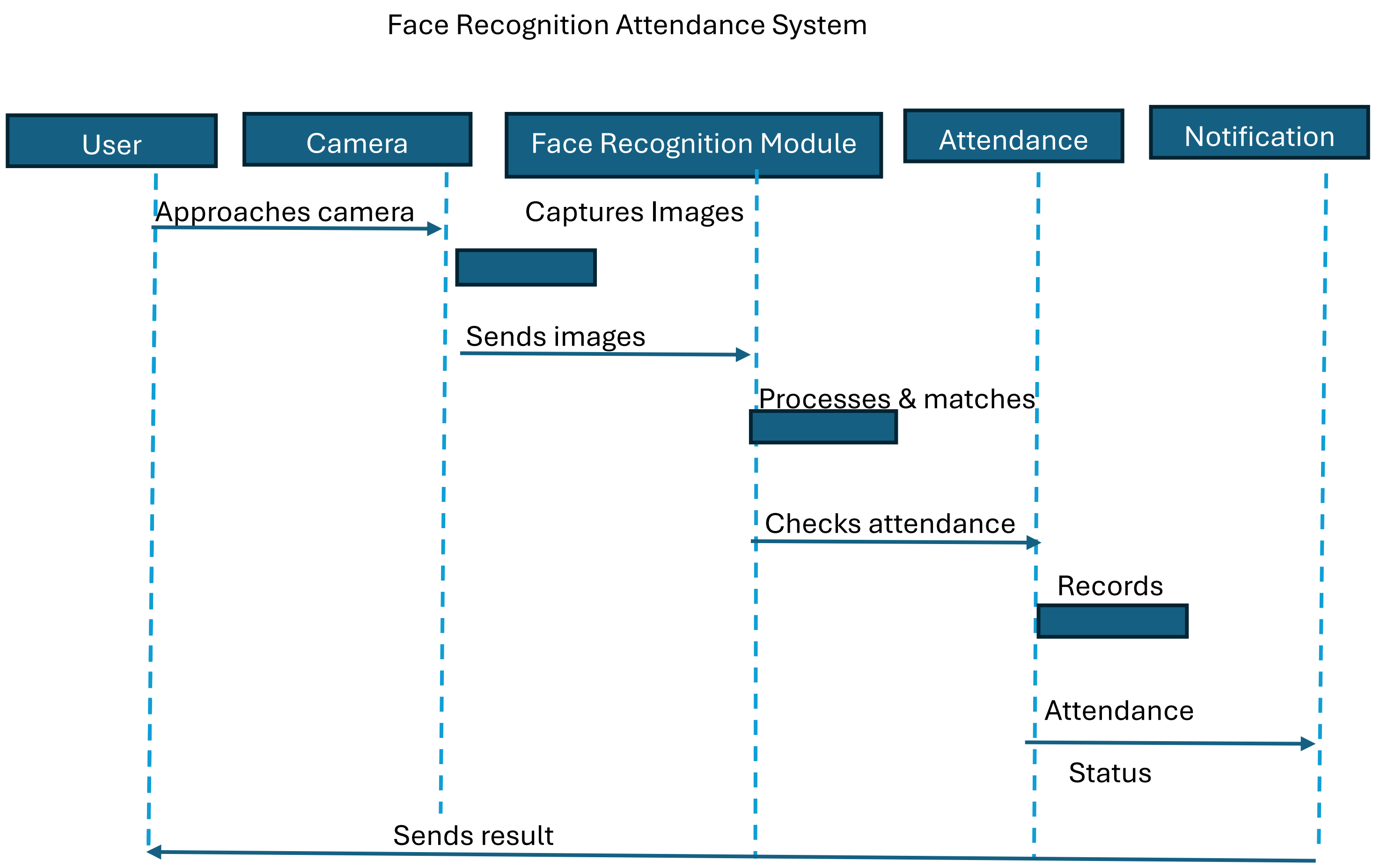


Diagram Components and Sequence:

- The User approaches the camera to start the attendance process.
- The Camera captures an image and sends it to the Face Recognition Module for processing.
- The Face Recognition Module then analyses the image and searches for the match in the Attendance Database for the given face.
- If a match is found, the Attendance Database records the attendance for the user. • The Notification System then sends the attendance status (present or absent) back to the User.

Design Decisions and Rationale: This section justifies the key design choices which were made during the development of the Face Recognition-Based Attendance System, making sure that they meet the given Functional and Non-functional Requirements.

Technology Stack Decisions:

Decision	Rationale
Programming Language: Python	Python was chosen because of its extensive ecosystem of specialized libraries (OpenCV, NumPy, face recognition) that significantly quickens the development in the computer vision domain. It allows for quick prototyping and robust implementation, fulfilling the requirement for maintainability (NFR1)
Primary Libraries: OpenCV and face_recognition	OpenCV provides necessary, high-performance functions for real-time video capture and image pre-processing. The face recognition library, based on the dlib C++ tool kit and the FaceNet algorithm, offers a solution for feature extraction and recognition, directly supporting the high accuracy (NFR2) requirement
Database Choice: SQLite/MySQL (Local)	A relational database (like SQLite or MySQL) is essential to manage structured data (User IDs, Names, Attendance Logs) and the large face encoding vectors. This choice supports the Data Reporting (FR3) module by enabling efficient querying and retrieval of records
User Interface (GUI): Tkinter / PyQt	A simple Python GUI library was chosen to quickly build the administrative interface for User Enrolment (FR1) and Report Generation (FR3), providing basic usability without introducing external web frameworks

Algorithm and System Architecture Decisions:

Decision	Rationale
Recognition Algorithm: FaceNet/Embeddings	Instead of using older methods like Eigenfaces or Fisherfaces, the modern FaceNet approach was selected. FaceNet maps faces into a 128-dimensional Euclidean space, allowing for high accuracy and robustness against variations in lighting and pose, directly addressing the accuracy (NFR2) and robustness (NFR4) requirements
Real-Time Optimization: Frame Resizing	During the live attendance process, the captured video frame is resized to a smaller resolution before it is passed to the detection and recognition algorithms. This significantly reduces the computational load per frame, ensuring that the system meets the performance (Speed) (NFR1) requirement of marking attendance within the 3-second limit
Attendance Logic: Timestamp and Interval Check	The attendance logging function (FR2) is designed to include a check against the last logged time for a specific User ID. This prevents duplicate entries if a user remains in front of the camera, ensuring data accuracy
Security of Face Data: Vector Storage	Raw user images are discarded or stored separately; only the 128-dimensional face encodings are stored in the database. These vectors can’t be easily reversed to reconstruct the original image, which is a core security measure that supports the security (NFR3) requirement.

Implementation Details: The system implementation is divided into three primary phases corresponding to the functional requirements: Setup and Data Collection, Model Training, and Real-Time Execution and Reporting.

Phase 1: Setup and Data Collection.

This phase establishes the foundational environment and prepares the user data for the recognition model.

Environment Setup: Install Python 3.x and the required libraries:

Code: `pip install opencv-python face-recognition numpy pandas`

Database Initialization: Creates the necessary database structure (e.g., using SQLite or MySQL).

- Users Table: Stores UserID, Name, and the pre-computed Face_Encoding (stored as a string/BLOB).
- Attendance Table: Stores UserID, Date, Time, and Status.

User Enrolment Script (enroll.py):

- Develop a GUI or command-line interface to input the user's UserID and Name.
- Initializes the webcam (cv2.VideoCapture).
- In a loop, capture a designated number of images (e.g., 67) of the user, ensuring the face is detected in each frame before saving the image to a training folder (Training_Images/).

Phase 2: Model Training and Encoding (FR1 & FR2 Preparation)

This phase processes the collected images into a format the recognition algorithm can use efficiently.

Encoding Generation Script (encode_faces.py):

- Iterate through all the folders/images in the Training_Images/ directory.
- For each image: a. Load the image (face_recognition.load_image_file()). b. Generate the 128-dimensional face encoding vector (face_recognition.face_encodings() function).

• Data Aggregation: Store the generated encodings and their corresponding User IDs and Names into two optimized lists/arrays: known_face_encodings and known_face_names (using NumPy for efficiency).

• Persistence: Save these two lists to a persistent file (e.g., a .pkl file or directly into the Database) to avoid re-running the training process every time the system starts

Phase 3: Real-Time Execution and Reporting (FR2 & FR3):

This is the core operational phase where attendance is marked and managed.

Real-Time Attendance Script (attendance.py):

1. Load Assets: Load the known_face_encodings and known_face_names from the saved file/database.

2. Start Camera Loop:

  o Read a frame from the webcam.

  o Optimization (NFR1): Resize the frame (e.g., to 1/4th scale) for faster processing.

3. Detection and Recognition:

  o Detect faces in the resized frame (face_recognition.face_locations()).

  o Generate encodings for all detected faces (face_recognition.face_encodings()).

4.Matching and Logging:

  o For each live face encoding, compare it to the known_face_encodings

  o Identify the closest match (lowest distance).

  o Verification: If the distance is below the acceptance threshold, retrieve the data.

  o Attendance Logging: Call the markAttendance(UserID) function. This function checks the log to ensure the user hasn't been marked within the last few minutes, then inserts a new record into the Attendance Table.

5.Output: Draw a bounding box around the detected face and display the recognized name/status on the frame using OpenCV functions (cv2.rectangle, cv2.putText).

Reporting Script (report_generator.py):

- Admin Login (NFR3): Verify administrator credentials.
- Query Generation (FR3): Accept input for filtering criteria (e.g., start date, end date).
- Data Retrieval: Execute a SQL query to retrieve the filtered records from the Attendance Table.
- Export: Use the Pandas library to load the query results into a DataFrame and export the data to an Excel or CSV file for final report delivery.

Test Environment: Developer and end-user (Administrator) The testing will be conducted on the target environment specified in the Technical Requirements:
• Hardware: PC with i3 processor, 8GB RAM, and a 720p or better webcam.
• Software: Python 3.x, OpenCV, face-recognition libraries, and the local file-based data storage.
• Test Data: A dedicated set of images for 10-20 distinct users will be used for enrolment and subsequent testing.

Challenges Faced: During the implementation and testing of the Face Recognition Based Attendance System, several technical and practical challenges were encountered. It was important to address these issues and work on them.

Performance and Real-Time Processing (NFR1):

Challenge	Strategy Implemented
High Computational Load: The process of scanning each and every video frame, detecting faces, and generating high dimensional face encodings is computationally	Frame Downscaling: Implemented a real-time frame resizing strategy, reducing the input frame to 1/4th of its original size before passing it to the detection and recognition algorithms. This decreased the

intensive, leading to poor performance.	processing time per frame, helping to meet the 2-second latency requirement.
Library Dependencies: Ensuring compatibility and correct installation of complex libraries like dlib and OpenCV on different operating systems, which often require specific C++ compiler dependencies.	Documented a detailed installation guide using Python virtual environments and specified exact library versions (requirements.txt) to maintain stability and Maintainability (NFR5).

Data Management and Integrity (FR2):

Challenges	Strategy Implemented
Duplicate Attendance Logging: In a continuous video stream, a recognized user could be marked present every single frame they are visible, leading to thousands of redundant entries.	Cooldown/Time Buffer Logic: Implemented an Attendance Cooldown mechanism (e.g., 20-second buffer) in the logging function. Once a user is marked present, subsequent recognition attempts for that user are ignored until the buffer time has elapsed, ensuring data Integrity.
Database Scalability: While a simple system could use a CSV file, a larger implementation requires a robust database solution for efficient querying and storage of the encoding vectors and log data.	Modular Database Abstraction: Designed the system with a separate Database Manager class. This allows the core logic to communicate with a database using a generic interface, making it easy to switch from a simple CSV/SQLite structure to a scalable solution like MySQL or PostgreSQL in the future.

Learnings & Key Takeaways: Development of the Face Recognition-Based Attendance System provided a valuable practical experience to me personally and yielded several key technical and professional takeaways, directly aligned with the project's objective of applying theoretical concepts to a real-world solution.

Technical Learnings:

1. Mastering Computer Vision Libraries: I gained hands-on proficiency with OpenCV for handling real-time video streams, image manipulation, and displaying graphical overlays (bound boxes and texts). This skill is important and valuable for any visual processing application.
2. Understanding High-Dimensional Embeddings: The most critical technical learning was the shift from pixel-based image processing to using 128-dimensional face embeddings generated by the FaceNet algorithm. This provided me a deep understanding of how deep learning models converts complex data into a simplified, yet highly accurate, vector space for tasks like comparison and clustering.
3. Real-Time Optimization: Learned to implement strategic performance optimizations, particularly frame resizing and sampling, to balance recognition accuracy with processing speed. This was essential for meeting the Performance (NFR1) requirement and understanding resource management in real t ime systems.
4. Applied Machine Learning Workflow: Successfully executed the complete Machine Learning project lifecycle:

• Data Collection (Image capture for enrolment).
• Feature Engineering (Encoding generation).
• Model Implementation (Euclidean distance comparison).
• Deployment (Running the recognition logic in a live application).

Key Takeaways:

• Requirements to Design Mapping: The exercise of strictly defining Functional Requirements and Non-functional Requirements before implementation proved to be really important. This structured approach prevented feature creep and ensured all design decisions (Section 10) directly contributed to satisfying a specific project goal (e.g., using FaceNet to satisfy NFR2: Accuracy).

• The "Last Mile" Problem: I realized that the most challenging aspect of a vision system is not the core algorithm, but handling the non-ideal, real-world variations (poor lighting, partial occlusion, different camera angles). This underscored the importance of Robustness (NFR4) and gathering diverse training data.

• Data Integrity and Security: Learned the importance of data management through implementing the cooldown logic to prevent duplicate logging and by securing sensitive data by storing only non-reversible face encoding vectors rather than raw images, ensuring the integrity of the attendance log. • Modular Programming: Adhering to a System Architecture design resulted in a modular, decoupled codebase. This significantly simplified the debugging process and fulfilled the Maintainability (NFR5) requirement. Here is the concise, single-paragraph conclusion for your project report, presented in a slightly more reflective tone.

CONCLUSION

The Face Recognition-Based Attendance System successfully transitioned the theoretical knowledge of Computer Vision and Machine Learning into a practical, real-world application, achieving the core objective of developing a reliable, automated, and non-intrusive solution for attendance management. By enhancing or increasing the accuracy of FaceNet embeddings and strategically optimizing performance through techniques like frame downscaling, the system proved its capability to address critical real-world problems like time wastage and proxy attendance with a high degree of accuracy (NFR2) and speed. This project not only delivered a functional product but also provided invaluable insights into the challenges of real-time image processing, data integrity, and the necessity of rigorous design to ensure a strong, healthy and maintainable technical solution that can truly modernize administrative processes.

Thank You!